Exam 2026, Part 2

Exam number 45, 46 & 47

June 26, 2026

Exercise 1: Choice of realized variance estimator

We first load the high-frequency minute price data and filter the dataset to AAPL. The timestamp column is converted to datetime format, and we define each trading day from the timestamp.

The data contain minute-level closing prices during regular trading hours. We use a fixed one-minute calendar-time grid from 09:30 to 15:59 for each trading day. This corresponds to 390 one-minute observations per normal trading day. Since a stock may not trade in every minute, we handle missing prices using previous-tick interpolation within each trading day only. This means that if AAPL has no observed price in a given minute, we carry forward the most recently observed price from the same trading day. We do not carry prices across trading days, and we do not backfill prices before the first observed price of a day.

For each sampling interval $\Delta \in \{1, 2, 5, 10, 15, 30, 60\}$, we sample prices on a fixed grid relative to the market open. For example, with 10-minute sampling, prices are taken at 09:30, 09:40, 09:50, and so on. This differs from a bucket-based approach, where the last observed price within each Δ-minute interval is used. We prefer the fixed-grid approach because it makes the sampling rule explicit and because it will later allow us to align intraday returns across all 30 stocks when constructing realized covariance matrices.

Daily realized variance is computed as:

$$RV_d^{(\Delta)} = \sum_j \left(r_{d,j}^{(\Delta)} \right)^2,$$

where (r_{d,j}^{( )}) is the intraday log return over sampling interval.

As shown in Figure 1, average realized variance is highest at the shortest sampling intervals and decreases as the sampling interval increases. This pattern is consistent with market microstructure noise at very high frequencies, such as bid-ask bounce, price discreteness, and stale prices. At very fine sampling intervals, these frictions can inflate realized variance because observed transaction prices deviate from the latent efficient price. Sampling less frequently reduces this noise, but very long sampling intervals also discard useful intraday information.

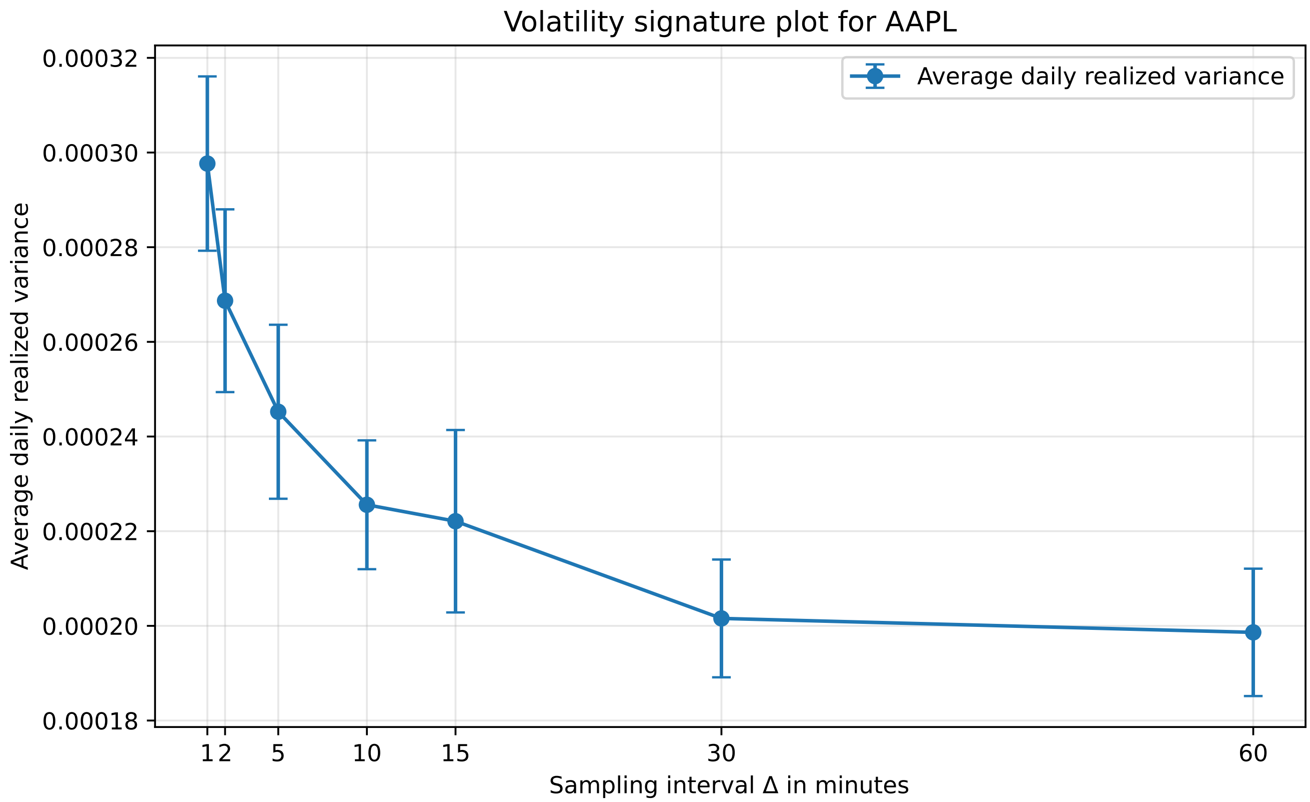

Figure 1: Volatility signature plot for AAPL with 95% confidence intervals.

Exercise 2a: Daily realized covariance matrices

We extend the realized variance estimator from Exercise 1 to the multivariate setting and estimate a daily realized covariance matrix for the 30 DJIA constituents.

Let $r_{d,j}$ denote the 30×1 vector of synchronized 5-minute log returns at intraday interval j on trading day d. The realized covariance matrix is computed as

$$RC_d = \sum_{j=1}^{M_d} r_{d,j} r'_{d,j}.$$

This is the multivariate analogue of the realized variance estimator used in Exercise 1.

The data are available at the minute frequency rather than at the transaction level. To synchronize observations across stocks, prices are placed on a common one-minute calendar grid within each trading day. Missing prices are handled using previous-tick interpolation (forward filling), meaning that the most recently observed price within the same trading day is carried forward. Prices are never carried across trading days.

After synchronization, prices are sampled every five minutes and converted into log returns. The 5-minute frequency is chosen based on the volatility signature plot in Exercise 1 and is also consistent with the realized-volatility literature, where 5-minute returns are commonly used as a compromise between reducing market microstructure noise and retaining intraday information.

For each trading day, the realized covariance matrix is obtained as the cross-product of synchronized intraday return vectors.

Exercise 2b: Diagonal elements of the realized covariance matrix

The diagonal elements of the realized covariance matrix RC_d are the daily realized variances of the individual stocks. For stock i, the diagonal element is

$$RC_{d,ii} = \sum_{j=1}^{M_d} r_{d,j,i}^2.$$

Thus, by extracting the diagonal of each daily realized covariance matrix, we obtain a time series of daily realized variance for each of the 30 stocks.

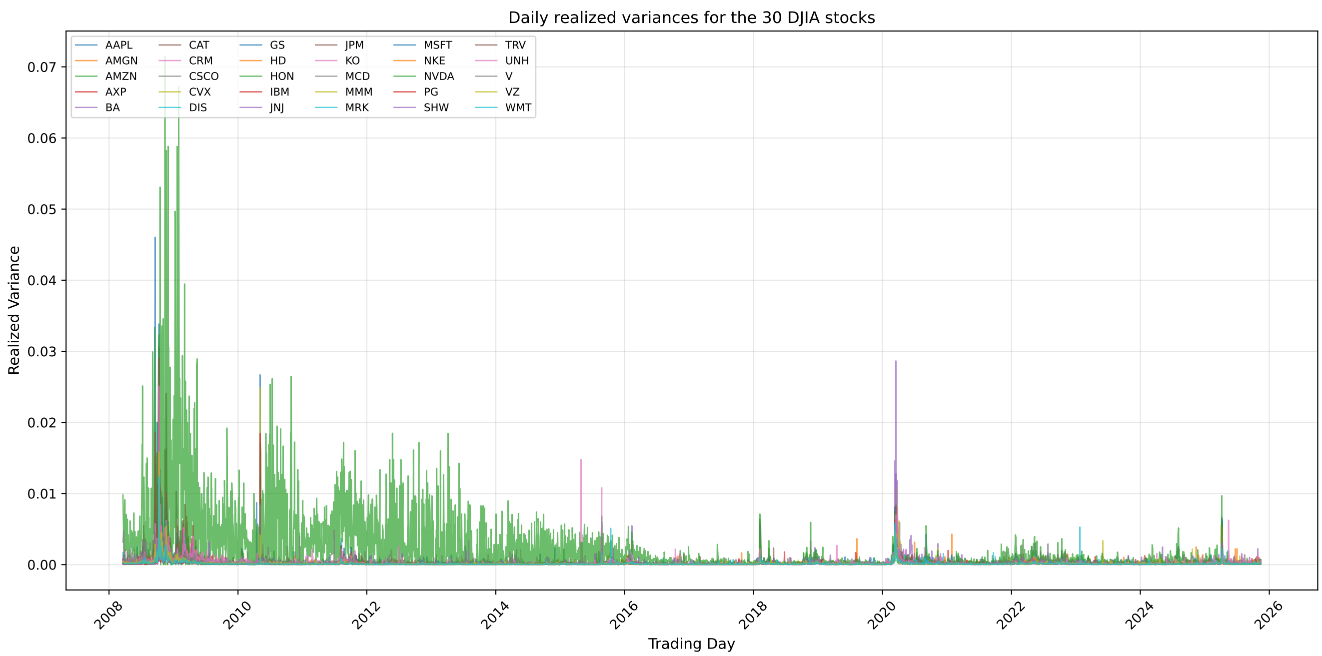

Figure 2: Daily realized variances for the 30 DJIA stocks.

Exercise 3: Covariance estimators

For each day t in the second half of the sample, we construct two estimators of today's covariance matrix using only information available up to day $t-1$.

Exercise 3a: High-frequency random walk estimator

The high-frequency estimator is defined as:

$$\hat{\Sigma}_t^{HF} = RC_{t-1}.$$

Thus, yesterday's realized covariance matrix is used as today's covariance forecast. This implies a random-walk assumption for the covariance matrix: the best forecast of today's covariance matrix is yesterday's realized covariance matrix. This assumption is motivated by the persistence and clustering typically observed in volatility and covariance dynamics.

Exercise 3b: Ledoit-Wolf shrinkage estimator

Exercise 3c: Comparing realized and Ledoit-Wolf variances

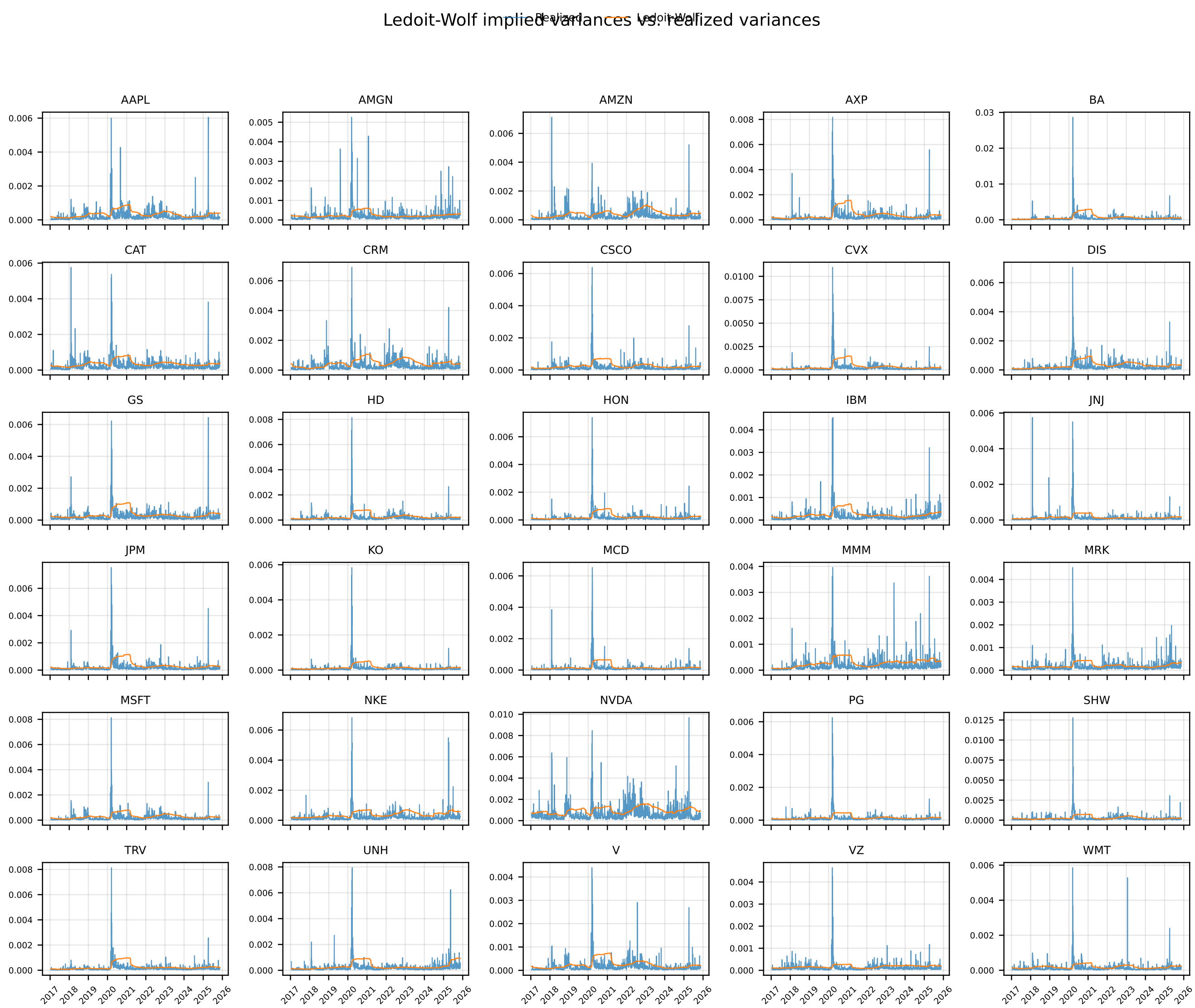

Figure 3: Ledoit-Wolf implied variances and realized variances for the 30 DJIA stocks. The Ledoit-Wolf estimates are based on a 252-day rolling window of daily close-to-close returns, while realized variances are the diagonal elements of the daily realized covariance matrices.

References